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SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

数字逻辑 DIGITAL LOGIC

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Lecture 3: Gate-Level Minimization

Today's Agenda



- Recap
- Context
 - Gate level minimization using the Map Method
 - Product of sums simplification
 - Don't Care Conditions
- Reading: Textbook, Chapter 3.1-3.5

Recap



- Boolean Algebra
 - DeMorgan's Law: $(x+y)'=x'y'$, $(xy)'=x'+y'$
 - Distributivity: $x(y+z)=xy+xz$; $x+yz=(x+y)(x+z)$
 - Complements: $x+x'=1$, $x \cdot x'=0$
 - etc
- Boolean Functions
 - Literals, Product term, Sum term
 - Unique: Truth table (2^n entries for n variables)
 - Not unique: Algebraic forms (simplifiable to reduce literals/terms for cheaper circuits).
- Canonical & Standard Forms
 - Canonical (unique)
 - Sum of Minterms (som: $\sum m_i$, all variables)
 - Product of Maxterms (pom: $\prod M_i$, $M_i=m_i'$)
 - Standard (not unique)
 - Sum of Products (SOP)
 - Product of Sums (POS)
- Other Logic Operations
 - Multi-input extension: AND/OR/XOR are associative; NAND/NOR are not.

Outline



- **Map Method Simplification**
- Product of sums simplification
- Don't Care Conditions



Boolean function simplification

- A function's truth-table representation is unique, while its algebraic expression is not unique.(i.e. sop and pos)
- Complexity of digital circuit (gate count) $\propto$ complexity of algebraic expression (literal count)
 - $F=x'y'z+x'yz+xy'$ (3 Product/AND term, 8 literals)
 - $F=x'z+xy'$ (2 AND terms, 4 literals)
- The simplest algebraic expression is one that has minimum number of terms with the smallest possible number of literals in each term
- Methods for gate-level minimization:
 - **Algebraic method**(逻辑代数): Boolean algebra (Last lecture)
 - **Karnaugh map**(卡诺图): the map method (This lecture)



Karnaugh Map (K-map)

- An array of squares each representing one minterm to be minimized
- Each K-map defines a unique Boolean function
 - A Boolean function can be represented by a truth table, a Boolean expression, or a map
- K-map is a visual diagram of all possible ways a function may be expressed
- Used for manual minimization of Boolean functions

Merging Minterms

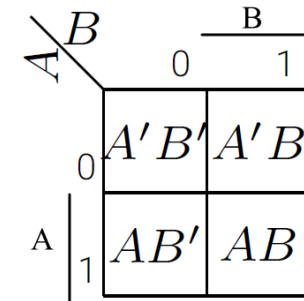
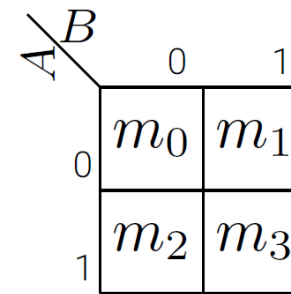
- In function F , m_1 and m_3 in the truth table differ only in one position
 - $001 \rightarrow x'y'z$
 - $011 \rightarrow x'yz$
 - $0?1$, with ? matching either 0 or 1, meaning the value of input y does not affect the function's output.
- The minterms in a function can be merged to form a simpler product term
 - $F = x'y'z + x'yz = x'z(y' + y) = x'z$

x	y	z	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

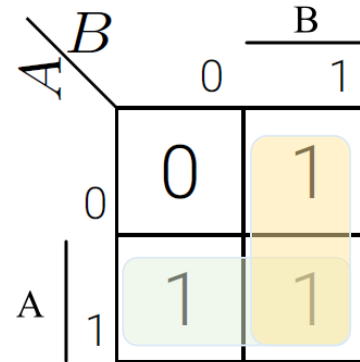
Two-Variable Map



- A two-variable map
 - is A truth table in square diagram
 - 4 minterms: $A'B'$, $A'B$, AB' , AB
 - row 0 stands for A' ; row 1 stands for A
 - column 0 stands for B' ; column 1 stands for B



	A	B	F
m_0	0	0	0
m_1	0	1	1
m_2	1	0	1
m_3	1	1	1



It's ok for groups to overlap,
if that makes them larger

$$m_1 + m_2 + m_3 = A'B + AB' + AB = A + B$$

Three-variable Map

- Minterms are arranged in the Gray-code sequence
- Any 2 (horizontally or vertically) adjacent squares differ by exactly 1 variable, which is complemented in one square and uncomplemented in the other.
- Any 2 minterms in adjacent squares that are ORed together will cause a removal of the different variable (adjacent applies not only the middle squares but also the boundary squares)

		BC			
		00	01	11	10
A	0	m_0	m_1	m_3	m_2
	1	m_4	m_5	m_7	m_6

Three-variable Map

- Example (adjacent squares)
- $m_5 + m_7 = AB'C + ABC = AC(B + B') = AC$
- $m_0 + m_2 = A'B'C' + A'BC' = A'C'(B + B') = A'C'$
- $m_4 + m_6 = AB'C' + ABC' = AC'(B' + B) = AC'$
- $m_1 + m_3 + m_5 + m_7$
 $= A'B'C + A'BC + AB'C + ABC = A'C(B + B') + AC(B + B')$
 $= A'C + AC = C$

		B			
		00	01	11	10
A	0	m_0	m_1	m_3	m_2
	1	m_4	m_5	m_7	m_6

C

		B			
		00	01	11	10
A	0	m_0	m_1	m_3	m_2
	1	m_4	m_5	m_7	m_6

C

Example



- Simplify the following Boolean functions.

$$F = A'BC + A'BC' + AB'C' + AB'C$$
$$= A'B + AB'$$

$\begin{array}{c} BC \\ \swarrow \end{array}$		$\begin{array}{c} B \\ \hline \end{array}$			
		00	01	11	10
$\begin{array}{c} A \\ \hline \end{array}$	0	0	0	1	1
	1	1	1	0	0

$\underbrace{\hspace{10em}}_C$

Green circle: $A'BC + A'BC' = A'B$
Red circle: $AB'C' + AB'C = AB'$

$$F = A'BC + AB'C' + ABC' + ABC$$
$$= BC + AC'$$

$\begin{array}{c} BC \\ \swarrow \end{array}$		$\begin{array}{c} B \\ \hline \end{array}$			
		00	01	11	10
$\begin{array}{c} A \\ \hline \end{array}$	0	0	0	1	0
	1	1	0	1	1

$\underbrace{\hspace{10em}}_C$

Green circle: $A'BC + ABC = BC$
Red circle: $AB'C' + ABC' = AC'$

Example



- Simplify the following Boolean functions.

$$F = \sum(1, 2, 3, 5, 7) = C + A'B$$

		B			
		BC			
A	0	0	1	1	1
	1	0	1	1	0

Groups: A horizontal group of four 1s (A'B) and a vertical group of four 1s (C).

$$F = \sum(0, 2, 4, 5, 6) = C' + AB'$$

		B			
		BC			
A	0	1	0	0	1
	1	1	1	0	1

Groups: A horizontal group of four 1s (AB') and a vertical group of four 1s (C').

It's ok for groups to overlap,
if that makes them larger

Exercise



- Simplify the following Boolean function.
 - $F = A'C + A'B + AB'C + BC$
 $= ?$

Exercise



- Simplify the following Boolean function.
 - $F = A'C + A'B + AB'C + BC = ?$
- Solution:
 - Express it in sum of minterms.
 - Find the minimal sum of products expression.
 - $F = A'C + A'B + AB'C + BC$
 $= A'C(B+B') + A'B(C+C') + AB'C + (A+A')BC$
 $= A'BC + A'B'C + A'BG + A'BC' + AB'C + ABC + A'BG$
 $= \sum(1, 2, 3, 5, 7) = C + A'B$

A Karnaugh map for the Boolean function $F = A'C + A'B + AB'C + BC$. The map is a 2x4 grid with variables A and B as row and column headers, and C as a sub-header for the columns. The cells are labeled with 0 or 1. The 1s are located at (A=0, B=0, C=1), (A=0, B=1, C=1), (A=0, B=1, C=0), (A=1, B=0, C=1), and (A=1, B=1, C=1). The cells are grouped into two prime implicants: a red group covering (A=0, B=0, C=1), (A=1, B=0, C=1), (A=0, B=1, C=1), and (A=1, B=1, C=1), which corresponds to the term C ; and a green group covering (A=0, B=1, C=1), (A=0, B=1, C=0), and (A=1, B=1, C=1), which corresponds to the term $A'B$. The intersection of these two groups is the cell (A=0, B=1, C=1).

		B			
		00	01	11	10
A	0	0	1	1	1
	1	0	1	1	0

Four-Variable Map



- The map
 - 16 minterms
 - Combinations of 2, 4, 8, and 16 adjacent squares

$AB \backslash CD$		C			
		00	01	11	10
A	00	m_0	m_1	m_3	m_2
	01	m_4	m_5	m_7	m_6
	11	m_{12}	m_{13}	m_{15}	m_{14}
	10	m_8	m_9	m_{11}	m_{10}

Implicants

- Implicant: any product term that implies the function
 - A product term that makes a function to be true
- Prime implicant (PI) (质蕴含)
 - A 1-product term obtained by combining the maximum possible number of adjacent squares in the map.
- Essential prime implicant (EPI) (基本质蕴含)
 - If a minterm in a square is covered by only one prime implicant

	minterm	implicant
m_1	√	√
m_2	√	X
$x'z$	X	√

1-minterm

0-minterm

1 product term

x	y	z	F	
0	0	0	0	m_0
0	0	1	1	m_1
0	1	0	0	m_2
0	1	1	1	m_3
1	0	0	1	m_4
1	0	1	1	m_5
1	1	0	0	m_6
1	1	1	0	m_7



Tips for simplification

- Simplification Steps:
 - Determine all essential prime implicants.
 - Find other prime implicants that cover remaining minterms.
 - Logical sum all prime implicants.
- Tips:
 1. Adjacent Coverage(相邻覆盖): Loop adjacent 1 terms, making each loop as large as possible.
 2. Minimal Loops(最少圈数): Use the fewest loops to cover all 1 terms for simplification.
 3. Avoid Redundancy(避免冗余): Ensure each loop covers at least one unique 1 not covered by other loops.
 4. Boundary Connectivity(边界连通): Treat the map as continuous—left/right and top/bottom edges are adjacent.
 5. Power-of-Two Priority(幂次优先): Prioritize looping groups of 2^n 1s (e.g., 8, 4, 2, 1) for maximum simplification.

Example



- Simplify the following Boolean functions

- $F = m_1 + m_5 + m_{10} + m_{11} + m_{12} + m_{13} + m_{15}$

- start with EPIs

- **Green circle:** $A'B'C'D + A'BC'D = A'C'D$

- **Purple circle:** $ABC'D' + ABC'D = ABC'$

- **Red circle:** ...

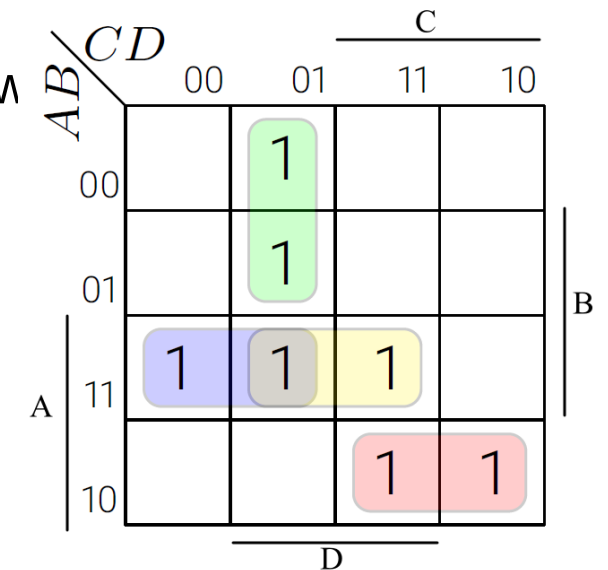
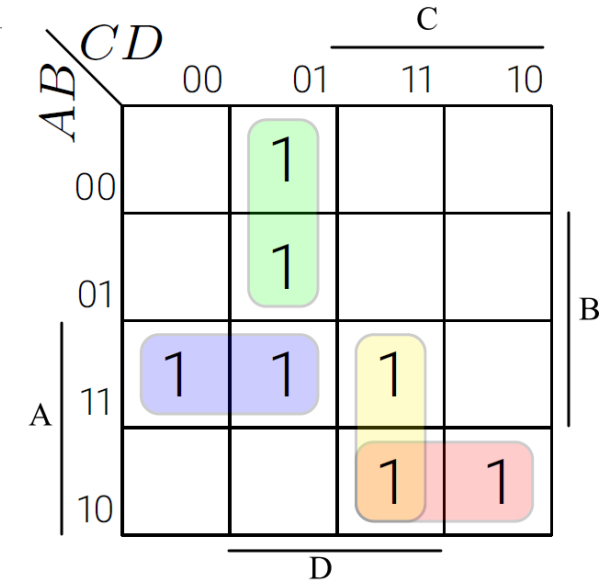
- ...

- $F = A'C'D + ABC' + ACD + AB'C$

- This reduced expression is not a unique one

- If pairs are formed in different ways, the simplified expression w

- $F = A'C'D + ABC' + ABD + AB'C$



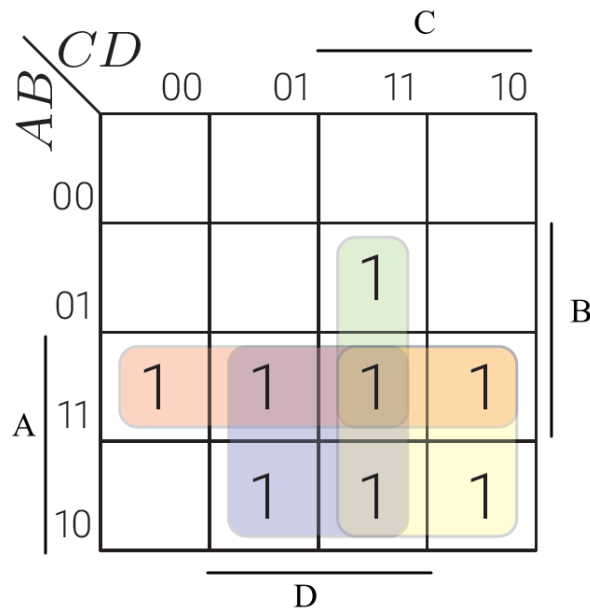
Example



- Simplify the following Boolean functions.

$$F(A,B,C,D) = \sum(7, 9, 10, 11, 12, 13, 14, 15)$$

$$F = AB + AC + AD + BCD$$



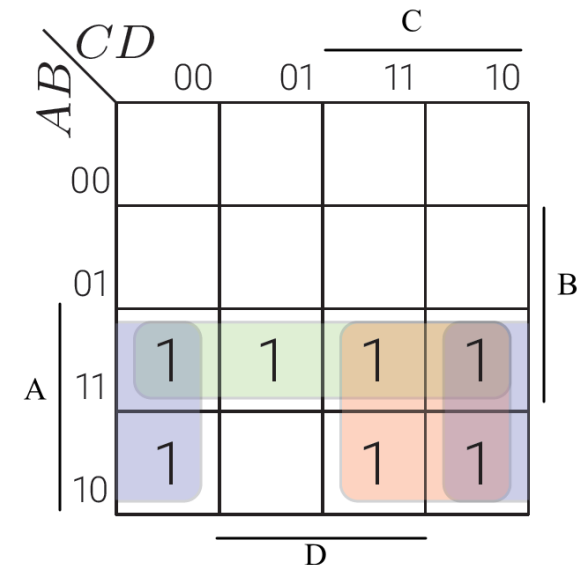
$$F(A,B,C,D) = ABCD + AB'C'D' + AB'C + AB$$

$$F = ABCD + AB'C'D' + AB'C(D + D') + AB(C + C')(D + D')$$

$$= \dots$$

$$= \sum(8, 10, 11, 12, 13, 14, 15)$$

$$= AB + AC + AD'$$



Exercise



- Simplify the following Boolean functions.

$$\begin{aligned} F(A,B,C,D) \\ &= \sum(0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14) \\ &= ? \end{aligned}$$

$$\begin{aligned} F(A,B,C,D) \\ &= A'B'C' + B'CD' + A'BCD' + AB'C' \\ &= ? \end{aligned}$$

Exercise



- Simplify the following Boolean functions.

$$\begin{aligned}
 F(A,B,C,D) &= \sum(0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14) \\
 &= C' + A'D' + BD'
 \end{aligned}$$

		C			
		00	01	11	10
A	00	1	1		1
	01	1	1		1
	11	1	1		1
	10	1	1		
		D			
		B			

$$\begin{aligned}
 F &= A'B'C' + B'CD' + A'BCD' + AB'C' \\
 &= A'B'C'(D + D') + B'CD'(A + A') + \\
 &\quad A'BCD' + AB'C'(D + D') \\
 &= A'B'C'D + A'B'C'D' + AB'CD' + A'B'CD' + \\
 &\quad A'BCD' + AB'C'D + AB'C'D' \\
 &= \sum(0, 1, 2, 6, 8, 9, 10) \\
 &= B'C' + B'D' + A'CD'
 \end{aligned}$$

		C			
		00	01	11	10
A	00	1	1		1
	01				1
	11				
	10	1	1		1
		D			
		B			

K-map Summary

- Any 2^k adjacent squares, $k=0,1,\dots,n$, in an n -variable map represent an area that gives a product term of $n-k$ literals

K	# of adjacent squares	# of literals left in a term in an n-variable map		
		n=2	n=3	n=4
0	1	2	3	4
1	2	1	2	3
2	4	0	1	2
3	8		0	1
4	16			0

- Five-Variable Map
 - Map for more than four variables becomes complicated
 - Five-variable map: two four-variable map (one on the top of the other), contains 2^5 or 32 cells.

Outline



- Map Method Simplification
- **Product of sums simplification**
- Don't Care Conditions



Product of Sums Simplification

- Previous Examples are Sum of Product Simplification
 - E.g. $F = AB + A'D + AB'C$ (Product of sum form)
- How to find Product of Sum simplification
 - E.g. $F = (A+B)(B+C')$ (Sum of Product form)
- POS simplification Steps
 - Simplified F' in the form of sum of products
 - Group adjacent 0-minterms squares together
 - Apply DeMorgan's theorem $F = (F')'$
 - F' : sum of products $\rightarrow$ F : product of sums

Example

- Simplify the Boolean function into product of sums form:

- $F(A,B,C,D) = \sum(2, 3, 7, 10, 11, 15)$

- Solution

- Step1: group the 0-minterms to find **F complement**

$$\begin{aligned} F' &= \sum(0, 1, 4, 5, 6, 8, 9, 12, 13, 14) \\ &= C' + BD' \quad (\text{Group 0 minterms}) \end{aligned}$$

- Step2: find the complement again to get original F

$$\begin{aligned} F &= (F')' = (C' + BD')' \\ &= C(B' + D) \quad (\text{DeMorgan's}) \end{aligned}$$

		C			
		00	01	11	10
A	00	0	0	1	1
	01	0	0	1	0
	11	0	0	1	0
	10	0	0	1	1
		D			
		B			

Exercise



- simplify $F(A, B, C, D) = \sum(0, 1, 2, 5, 8, 9, 10)$ into
 - sum-of-products form
 - $F = ?$
 - product-of-sums form
 - $F = ?$

Exercise

- simplify $F(A, B, C, D) = \sum(0, 1, 2, 5, 8, 9, 10)$ into
 - sum-of-products form
 - $F = B'D' + B'C' + A'C'D$ (Group 1-minterms)
 - product-of-sums form
 - $F' = AB + CD + BD'$ (Group 0-minterms)
 - $F = (A'+B')(C'+D')(B'+D)$ (DeMorgan's)

		CD		C		
		00	01	11	10	
A	AB	00	01	11	10	B
	00	1	1	0	1	
	01	0	1	0	0	
	11	0	0	0	0	
	10	1	1	0	1	
		D				

		CD		C		
		00	01	11	10	
A	AB	00	01	11	10	B
	00	1	1	0	1	
	01	0	1	0	0	
	11	0	0	0	0	
	10	1	1	0	1	
		D				

Outline



- Map Method Simplification
- Product of sums simplification
- **Don't Care Conditions**

Don't care conditions

- Incompletely specified functions
 - Functions that have unspecified outputs for some input combinations
 - E.g. output are unspecified for 1010 to 1111 in 4-bit BCD code
- Don't-care conditions
 - Unspecified minterms of a function, don't-cares, Xs
 - Can be used on a map to provide further simplifications of the Boolean expression
 - Each X can be assigned an arbitrary value, 0 or 1, to help simplification procedure

Example



- Simplify $F(A, B, C, D) = \sum(1, 3, 7, 11, 15)$ with don't-care conditions $d(A, B, C, D) = \sum(0, 2, 5)$.
 - $F = A'B' + CD$
 - or $F = A'D + CD$
 - Just make sure all 1 minterms are circled, thus both simplifications are acceptable

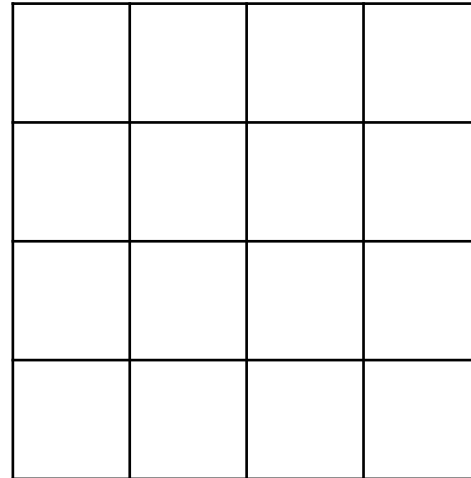
		C			
		00	01	11	10
A	00	X	1	1	X
	01		X	1	
	11			1	
	10			1	
		D			

		C			
		00	01	11	10
A	00	X	1	1	X
	01		X	1	
	11			1	
	10			1	
		D			

Exercise



- Using the Karnaugh map method obtain the minimal sum of the products expression for the function $F(A,B,C,D) = \Sigma(0, 2, 3, 6, 7) + d(8, 10, 11, 15)$



Exercise



- Using the Karnaugh map method obtain the minimal sum of the products expression for the function $F(A,B,C,D) = \Sigma(0, 2, 3, 6, 7) + d(8, 10, 11, 15)$
- $F = A'C + B'D'$

		CD				
		00	01	11	10	
A	AB	00	01	11	10	B
	00	1	0	1	1	
	01	0	0	1	1	
	11	0	0	X	0	
	10	X	0	X	X	D